MIL-S-17709C(GL)
28 October 1977
SUPERSEDING
MIL-S-17709B
27 August 1964

MILITARY SPECIFICATION

SLOTTING TOOL, COMMUTATOR, HAND: AND

SLOTTER AND SCRAPER, COMMUTATOR, HAND

This specification is approved for use by the US Army Natick Research and Development Command (GL), Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hand tools for use in slotting and scraping slots in commutators of electric motors and generators.

1.2 Classification. Tools shall be furnished in the following types as specified (see 6.2):

Type D - Slotting tool
Type E - Slotter and scraper

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5110

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STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The slotting tools and the slotter and scrapers shall be the manufacturer's standard product except for changes necessary to comply with this specification.

3.2 Materials (see 6.3). Materials not definitely specified shall be of the quality normally used by the manufacturer for his current commercial product.

3.3 Design and construction.

3.3.1 Handles. The type D slotting tool and type E slotter and scraper handles shall be wood, metal or plastic shaped to provide a good hand grip. Plastic handles, where furnished, shall be unbreakable under ordinary service conditions, and shall show no evidence of chipping, cracking or distortion when tested as specified in 4.3.1.

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3.3.2 Type D, slotting tool. Slotting tools shall be in general conformance with figure 1 with an overall length of 10 inches (254 mm) plus or minus 1 inch (25.4 mm). The working end shall be single end, draw cut files of high grade steel.

3.3.3 Type E, slotters and scrapers. Slotters and scrapers shall be in general conformance with figure 2 with a minimum overall length of 6-1/4 inches (158.7 mm). The slotters and scrapers shall have adjustable heads permitting rotation of the removable cutting bit, contained therein, to any angle. The cutting bit, hardened to Rockwell C 60-63 (see 4.3.2) shall be sharpened for "U" cutting on one end and "V" cutting on the other.

3.4 Finish. The tools shall be finished in accordance with the manufacturer's commercial practice.

3.5 Marking for identification. The tools shall be permanently and legibly marked with the manufacturer's name or readily identifiable trademark in accordance with the manufacturer's commercial practice.

3.6 Workmanship. The tools shall be free of rust, burrs, sharp edges (except cutting edges) bent or fractured components, wood handles, when applicable, shall be free of splinters or knots.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.2.2 End item inspection. A lot shall consist of all the tools of one type offered for inspection at one time. The sample unit shall be one completely fabricated slotting tool or slotter and scraper.

4.2.2.1 Visual examination. The end item shall be examined for the defects classified below. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not in accordance with manufacturer's commercial practice		X
Construction and workmanship	Component not of specified design	X	
	Component bent, fractured, malformed, out of alignment, or otherwise damaged	X	
	Sharp or burred edge or projection (except cutting edges and file teeth)	X	
	Evidence of rust	X	
	Wood handles not free of splinters or knots	X	
Identification marking	Missing, illegible, incorrect, incomplete, or not permanent type		X

4.2.2.2 Dimensional inspection. Inspection shall be made of the end item for compliance with the dimensions specified herein. Any deviation from the specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. Tools with plastic handles only shall be tested as specified in 4.3.1 and type E slotters and scrapers shall be tested as specified in 4.3.2. Any nonconformance shall constitute a defect. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

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4.2.3 Inspection of preparation for delivery. Preservation-packaging, packing and marking shall be inspected in accordance with the quality assurance provisions of PPP-P-40.

4.3 Tests.

4.3.1 Impact test for tools with plastic handles. The tool shall be held securely as close to the handle as possible with the handle free from restraint. An impact equivalent to 5 foot pounds (6.78 joules) shall be delivered to the handle perpendicular to the long axis of the tool, at or near the middle of the natural grip, by a metal bar with a "vee" edge. This test shall be repeated five times. The handle shall be examined for compliance with the requirements specified in 3.3.1 and any evidence of nonconformance shall constitute failure of this test.

4.3.2 Hardness test. The cutter bit of the type E slotter and scraper shall be tested in accordance with ASTM E 18. A minimum of three determinations shall be made on each sample. Nonconformance to the requirements specified in 3.3.3 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging, packing, and marking. Preservation-packaging, packing, and marking for level A, B, or C shall be in accordance with the applicable provisions of PPP-P-40 (see 6.2).

6. NOTES

6.1 Intended use.

6.1.1 Type D. Type D slotting tools are intended for light or intermittent undercutting of the mica insulation between commutator bars.

6.1.2 Type E. Type E slotters and scrapers are intended for cleaning commutator slots and for slightly beveling the copper edges.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of preservation-packaging, packing and marking (see 5.1).

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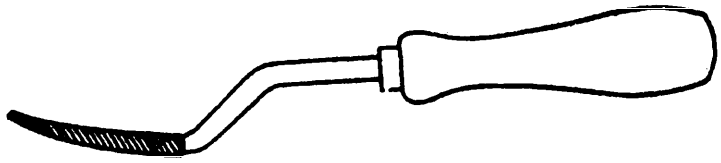


FIGURE 1. Slotting tool.

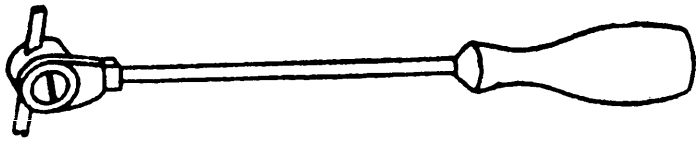


FIGURE 2. Slotter and scraper

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6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 6.3).

6.4 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-S-17709C(GL)
Slotting Tool, Commutator, Hand: And Slotter And Scraper, Commutator, Hand

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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